

ADITYA GIRISH

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EDUCATION

Bachelor of Technology (B.Tech.), Computer Science, 2022-2026

PES University, Bengaluru

PUBLICATIONS

When Does Structure Help? Graph vs. Tabular Models for Agent-Workflow Cost Prediction

AAAI Conference on Artificial Intelligence (AAAI 2027)

Under Review - Main Technical Track

AAAI Press Proceedings · CORE A* · Double-blind peer review

Solo Author

Establishes when graph neural networks beat tabular models for agent-workflow cost prediction, showing across 357K real coding-agent rounds that graph models lose when execution costs do not propagate through structure.

MSTRAE: Multi-Scale Temporal Recurrent Autoencoder for Flight Manoeuvre Recognition

Future Technologies Conference (FTC) 2026

Accepted - Oct 2026, Berlin

Springer Lecture Notes in Networks and Systems · Indexed: Web of Science, SCOPUS, INSPEC

Lead Author

Achieves zero-shot flight manoeuvre recognition on Cessna 172 telemetry with a multi-scale LSTM autoencoder in PyTorch, HMM-based unsupervised clustering, and an LLM-based evaluation framework replacing hand-labelled ground truth.

EXPERIENCE

TruEstate

Bengaluru, India

AI/ML Engineer

Jul 2026 - Present

- Specified the full voice-AI stack for an outbound cold-calling pipeline — **telephony, STT, LLM, TTS** — setting per-hop latency budgets and barge-in behaviour so callers interrupt mid-utterance without losing turn state.
- Replaced static call scripts with **per-lead context curation** that joins CRM history and live listing data at call time, so each agent opens on the properties and prior interactions specific to that lead.
- Automated **Meta ad creative production** end to end, building a pipeline that renders image and video variants directly from structured listing fields instead of a per-listing design request.

Software Engineering Intern - AI

Jan 2026 - Jul 2026

- Collapsed a **three-agent design into one classification-routed agent** spanning sales, service-delivery, and post-sales, using tool calling for user-data retrieval and hierarchical vector search; slated for production.
- Cut broker site-build time from **2 hours to 5 minutes** and lifted lead quality **60%** with an agent-based CMS on GCP generating microsites in one click, with Cloudflare deploys, OTP verification, and CRM lead routing.
- Pitched AI capabilities to **500+ brokers across two industry events**, translating model behaviour into broker-level business value and gathering the requirements that shaped subsequent product iterations.
- Shipped an image-generation pipeline on **Gemini 2.5 Flash Image** that virtually staged interiors across **200+ property listings**, removing manual design turnaround from the listing-publish path.
- Originated and demoed a **2D-to-3D reconstruction pipeline** turning architectural floor plans into interactive **Three.js** models, using a **Gemini VLM** for layout-element detection and spatial parsing.

AI Backend Engineer Intern

Apr 2025 – May 2025

Viable Ideas Private Limited

Bengaluru, India

- Deployed an **LLM-based** inventory prediction model to production at **85% classification accuracy** on transaction data from 4 bank sources; automated multi-source ingestion cut manual workflows **70%** and cost **40%**.

PROJECTS

QRAG : Quantized Bi-Level Retrieval Augmented Generation

- Achieved **95.1% recall@10** on FIQA at a **3–4× memory reduction** by quantizing the coarse and fine retrieval tiers independently in a bi-level index built for memory-constrained retrieval.
- Beat **FAISS IVF-PQ by 21.5pp in recall** at comparable memory budgets across a 57,000-document benchmark, isolating where bi-level quantization wins over standard ANN index compression.

OriginScale : Low-Latency Deterministic Clustering Initialization

- Reduced clustering initialization from a **~3.5s** average to a constant **26ms** on datasets up to 200K entries, replacing the standard initialization's data-dependent sampling with a deterministic scheme.
- Delivered **1.68× lower latency** in a real-time food-delivery dispatch system with no loss in assignment quality, where response time rather than clustering objective is the binding constraint.